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System and method for generating a context-based user interface

Abstract

A system includes a memory and a processor configured to record a plurality of attributes associated with one or more hardware devices, one or more software applications, or a combination thereof used by the user to perform a plurality of interactions, wherein the plurality of attributes are indicative of one or more characteristics associated with the user. Based on the recorded attributes, the processor determines a user profile associated with the user. Subsequently, in response to detecting that the user has requested to perform an interaction in a UI environment, the processor obtains a UI model associated with the UI environment, obtains a set of conformance rules associated with the determined user profile of the user, and generates a UI screen for the UI environment based on the UI model and the set of conformance rules.

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Background/Summary

TECHNICAL FIELD

(1) The present disclosure relates generally to data processing, and more specifically to a system and method for generating a context-based user interface.

BACKGROUND

(2) When a user interface associated with a software application is designed, the user interface needs to satisfy several compliance standards including regulatory standards, accessibility standards (e.g., Americans with Disabilities Act, ADA standards), data privacy/security standards etc. Often different sets of standards apply to different types of users. Thus, a user interface presented to a particular user may need to conform to the set of standards defined for a type of the particular user. For example, accessibility standards may define different sets of UI design rules, wherein each set of UI design rules applies based on a type of disability associated with a particular user who is using the software application. Non-conformance to compliance standards may have several disadvantages including fines, data security breaches and loss of user confidence.

SUMMARY

(3) The system and method implemented by the system as disclosed in the present disclosure provide technical solutions to the technical problems discussed above by providing intelligent techniques for conformance of user interfaces (UIs) to defined standards.

(4) For example, the disclosed system and methods provide the practical application of intelligently generating UI screens associated with software applications based on characteristics of a user using the software application and a set of conformance rules defined for a type of the user. As described in embodiments of the present disclosure, based on monitoring a plurality of interactions performed by a user, a UI manager records a plurality of attributes associated with one or more hardware devices, one or more software applications, or a combination thereof used by the user to perform the plurality of interactions, wherein the plurality of attributes are indicative of one or more characteristics associated with the user. Based on the recorded attributes, UI manager determines a user profile associated with the user. Subsequently, in response to detecting that the user has requested to perform an interaction in a UI environment, the UI manager obtains a UI model associated with the UI environment, obtains a set of conformance rules associated with the determined user profile of the user, and generates a UI screen for the UI environment based on the UI model and the set of conformance rules. Generating the UI screen includes selecting one or more UI elements of the UI model based on the set of conformance rules, wherein each UI element conforms to one or more conformance rules. The UI renders the generated UI screen in the UI environment on a user device of the user.

(5) By intelligently generating UI screens that conform to defined conformance rules, the disclosed system and method save computing resources (e.g., processing and memory resources) which would otherwise be used to analyze several sets of conformance rules and individual generate and render UI screens that conform to the defined rules. Accordingly, the disclosed system and methods improve processing efficiency of computing nodes used to generate and render UT screens in a computing infrastructure. Also, the disclosed system and methods generally improve the technology associated with generating and rendering user interfaces on computers.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) For a more complete understanding of this disclosure, reference is now made to the following brief description, taken in connection with the accompanying drawings and detailed description,

wherein like reference numerals represent like parts.
(2) FIG. **1** is a schematic diagram of a system, in accordance with an embodiment of the present disclosure; and
(3) FIG. **2** illustrates a flowchart of an example method for generating a UI screen for a user, in accordance an embodiment of the present disclosure.

DETAILED DESCRIPTION

(4) FIG. **1** is a schematic diagram of a system **100**, in accordance with certain aspects of the present disclosure. As shown, system **100** includes a computing infrastructure **102** including a plurality of computing nodes **104** connected to a network **180**. Computing infrastructure **102** may include a plurality of hardware and software components. The hardware components may include, but are not limited to, computing nodes **104** such as desktop computers, smartphones, tablet computers, laptop computers, servers and data centers, virtual reality (VR) headsets, augmented reality (AR) glasses and other hardware devices such as printers, routers, hubs, switches, input devices, output devices, and memory devices all connected to the network **180**. Software components may include software applications **110** that are run by one or more of the computing nodes **104** including, but not limited to, operating systems, user interface applications, web applications, third party software, database management software, service management software, metaverse software and other customized software programs implementing particular functionalities. For example, software code relating to one or more software applications **110** may be stored in a memory device and one or more processors may process the software code to implement respective functionalities. In one embodiment, a software application **110** may be configured to operate in a particular user interface (e.g., UI) environment **108** such as a web environment or a virtual environment (e.g., metaverse environment). For example, when a software application **110** is a web application, the UI associated with the web application is a web UI configured to run in a web environment. On the other hand, when a software application **110** is a metaverse application, the UI associated with the metaverse application is a metaverse UI configured to run in a metaverse environment.
(5) One or more of the computing nodes **104** may be operated by a user **106**. For example, a computing node **104** may provide a user interface (e.g., web UI and/or metaverse UI) using which a user **106** may operate the computing node **104** to perform data interactions within the computing infrastructure **102**. For example, a user **106** may use a laptop computer to access a web application running on a server, wherein both the laptop computer and the server are part of the computing infrastructure **102**. Similarly a user **106** may use VR glasses to access a metaverse application running on a server, wherein both the VR glasses and the server are computing nodes **104** of the computing infrastructure **102**.
(6) In one embodiment, at least a first portion of the computing infrastructure **102** may be representative of an Information Technology (IT) infrastructure of an organization/entity. For example, a first portion of the computing nodes **104** and a first portion of the software applications **110** may be part of the IT infrastructure of the organization. In this context, a second portion of the computing infrastructure **102**, which is different from the first portion of the computing infrastructure **102**, may not be a part of the IT infrastructure of the organization. For example, a second portion of the computing nodes **104** may not be part of the IT infrastructure of the organization and may be operated by users **106** to access software applications **110** running on computing nodes **104** that are part of the first portion of the computing nodes **104** belonging to the IT infrastructure of the organization. In one example scenario, the first portion of the computing infrastructure **102** may belong to a provider of a social media application/platform which provides access to the social media application/platform (e.g., web application or metaverse application) running on servers within the IT infrastructure belonging to the social media provider. Users of the social media platform may access the social media application running on the provider's servers using external computing nodes **104** that are not part of the IT infrastructure of the social media provider.

(7) One or more computing nodes **104** of the computing infrastructure **102** may be representative of a computing system that hosts software applications which may be installed and run locally or may be used to access software applications running on a server (not shown). The computing system may include mobile computing systems including smart phones, tablet computers, laptop computers, or any other mobile computing devices or systems capable of running software applications and communicating with other devices. The computing system may also include non-mobile computing devices such as desktop computers or other non-mobile computing devices capable of running software applications and communicating with other devices. In certain embodiments, one or more of the computing nodes **104** may be representative of a server running one or more software applications to implement respective functionality (e.g., UI manager **140**) as described below. In certain embodiments, one or more of the computing nodes **104** may run a thin client software application where the processing is directed by the thin client but largely performed by a central entity such as a server (not shown).
(8) Network **180**, in general, may be a wide area network (WAN), a personal area network (PAN), a cellular network, or any other technology that allows devices to communicate electronically with other devices. In one or more embodiments, network **180** may be the Internet.
(9) One or more computing nodes **104** of the computing infrastructure **102** may implement a UI manager **140** which, as further described below, is responsible for generating and rendering UIs associated with software applications **110**. The UI manager **140** comprises a processor **192**, a memory **196**, and a network interface **194**. The UI manager **140** may be configured as shown in FIG. **1** or in any other suitable configuration.
(10) The processor **192** comprises one or more processors operably coupled to the memory **196**. The processor **192** is any electronic circuitry including, but not limited to, state machines, one or more central processing unit (CPU) chips, logic units, cores (e.g., a multi-core processor), field-programmable gate array (FPGAs), application specific integrated circuits (ASICs), or digital signal processors (DSPs). The processor **192** may be a programmable logic device, a microcontroller, a microprocessor, or any suitable combination of the preceding. The processor **192** is communicatively coupled to and in signal communication with the memory **196**. The one or more processors are configured to process data and may be implemented in hardware or software. For example, the processor **192** may be 8-bit, 16-bit, 32-bit, 64-bit or of any other suitable architecture. The processor **192** may include an arithmetic logic unit (ALU) for performing arithmetic and logic operations, processor registers that supply operands to the ALU and store the results of ALU operations, and a control unit that fetches instructions from memory and executes them by directing the coordinated operations of the ALU, registers and other components.
(11) The one or more processors are configured to implement various instructions, such as software instructions. For example, the one or more processors are configured to execute instructions (e.g., UI manager instructions **198**) to implement the UI manager **140**. In this way, processor **192** may be a special-purpose computer designed to implement the functions disclosed herein. In one or more embodiments, the UI manager **140** is implemented using logic units, FPGAs, ASICs, DSPs, or any other suitable hardware. The UI manager **140** is configured to operate as described with reference to FIG. **2**. For example, the processor **192** may be configured to perform at least a portion of the method **200** as described in FIG. **2**.
(12) The memory **196** comprises one or more non-transitory computer-readable mediums such as disks, tape drives, or solid-state drives, and may be used as an over-flow data storage device, to store programs when such programs are selected for execution, and to store instructions and data that are read during program execution. The memory **196** may be volatile or non-volatile and may comprise a read-only memory (ROM), random-access memory (RAM), ternary content-addressable memory (TCAM), dynamic random-access memory (DRAM), and static random-access memory (SRAM).
(13) The memory **196** is operable to store a machine learning (ML) model **142**, user profiles **144**,

attributes **146**, UI models **148**, UI elements **150**, conformance rules **152**, and the UI manager instructions **198**. The UI manager instructions **198** may include any suitable set of instructions, logic, rules, or code operable to execute the UI manager **140**.

(14) The network interface **194** is configured to enable wired and/or wireless communications. The network interface **194** is configured to communicate data between the UI manager **140** and other devices, systems, or domains (e.g., computing nodes **104** etc.). For example, the network interface **194** may comprise a Wi-Fi interface, a LAN interface, a WAN interface, a modem, a switch, or a router. The processor **192** is configured to send and receive data using the network interface **194**. The network interface **194** may be configured to use any suitable type of communication protocol as would be appreciated by one of ordinary skill in the art.

(15) It may be noted that each of the computing nodes **104** may be implemented like the UI manager **140** shown in FIG. **1**. For example, each of the computing nodes **104** may have a respective processor and a memory that stores data and instructions to perform operations discussed above.

(16) When a user interface associated with a software application **110** (e.g., web application, metaverse application etc.) is designed, the user interface needs to satisfy several compliance standards including regulatory standards, accessibility standards (e.g., Americans with Disabilities Act, ADA standards), data privacy/security standards etc. These standards may be defined by an entity managing the software application **110** or by a government institution. Often different sets of standards apply to different types of users **106**. Thus, a user interface presented to a particular user **106** may need to conform to the set of standards defined for a type of the particular user. For example, accessibility standards may define different sets of UI design rules, wherein each set of UI design rules applies based on a type of disability associated with a particular user who is using the software application. For example, a UI presented to a color-blind user may need to adhere to a particular color scheme. On the other hand, a UI presented to another user with limited use of hands may need to provide voice to text features. In another example, different sets of data security/privacy rules may apply to corporate users and individual users of the same software application **110**. For example, computing nodes **104** used by individual users generally do not implement high data security protocols and thus are vulnerable to cyber-attacks. Accordingly, a UI presented to an individual user needs to include security features such as multi-factor authentication, face recognition, etc. On the other hand, computing systems used by corporate users typically implement high data security protocols. Thus, a UI presented to a corporate user may implement a more relaxed set of data security rules as compared to the rules applied for individual users. Non-conformance of a software application **110** to compliance standards may have several disadvantages including fines, data security breaches and loss of user confidence.

(17) Embodiments of the present disclosure describe techniques for generating UI screens **170** associated with a software application **110** based on characteristics of the user **106** who is using the software application **110**.

(18) In this context, UI manager **140** may be configured to determine a user profile **144** associated with a user **106** based on interactions **112** performed by the user **106** in the computing infrastructure **102**. Each user profile **144** may be associated with a particular type of users **106**. For example, a user profile **144** may be associated with users having a particular type of disability. In this example, user profiles **144** may include, but are not limited to, vision impaired profile, cognitive disability profile, seizure profile, attention deficit hyperactivity disorder (ADHD) profile, blind profile, and impaired motor skills profile. Other example user profiles **144** may include corporate user profile and individual user profile. UI manager **140** may be configured to determine a user profile **144** associated with a particular user **106** based by monitoring a plurality of interactions **112** performed by the user **106** in the computing infrastructure **102**. An interaction performed by a user **106** may include the user **106** accessing a software application **110** (e.g., a web application, metaverse application etc.) to perform a particular task such as posting on a social

media platform, sending an email message, transfer data files to other users or entities, purchase event tickets and the like. For example, the user **106** may log into a social media platform running on a server as a web application or a virtual application to consume content or to submit content.

(19) Monitoring interactions **112** performed by the particular user **106** may include recording a plurality of attributes **146** associated with one or more hardware devices, one or more software applications, or a combination thereof used by the user **106** while performing the monitored interactions **112**. The attributes **146** associated with interactions **112** monitored for the user **106** may indicate one or more characteristics associated with the user **106**. In one embodiment, attributes **146** may include a type of one or more hardware devices (e.g., input devices, output devices etc.), a type of one or more software applications, or a combination thereof used by the user **106** while performing the monitored interactions **112**. In this context, the type of hardware devices and/or software applications used by a particular user **106** may indicative of a particular type of disability associated with the user **106**, and thus may indicate a particular user profile **144** associated with the particular disability. For example, a user **106** having a particular type of disability may use specific hardware devices and/or specific software tools that are designed to help the users **106** having the particular type of disability to perform interactions **112** in the computing infrastructure **102**. For example, a vision impaired user may use a braille keyboard and/or a particular software plugin that converts voice to text and vice versa to enable the user **106** to issue voice commands. Other examples of assistive hardware devices and software tools that may be used by users **106** with different types of disabilities may include, but are not limited to, screen readers, adaptive keyboards, alternative input devices, braille displays, electronic screen magnifiers, augmentative and alternative communication software, dictation software, optical character recognition (OCR) software, screen magnification software and text to speech software. Each of these assistive hardware devices and software tools help users **106** with specific types of disabilities and thus are indicative of the type of disability associated with the user **106** using such devices and software.

(20) In another example, attributes **146** may indicate whether a user **106** is employed by a large organization or entity. For example, if a user **106** uses a high-end computer and/or uses an operating system or software tools typically used by large corporations, this is indicative of the user **106** being part of a large organization or entity. On the other hand, when a user **106** uses a low-end computer and/or uses home versions or student versions of an operating system of other software tools, this is indicative of the user **106** a low-end user.

(21) In one or more embodiments, UI manager **140** may be configured to monitor one or more computing nodes **104** used by a user **106** to perform a plurality of interactions **112** and identify the particular hardware devices and software applications used on the computing nodes **104** to perform the interactions **112**. Based on monitoring a plurality of interactions **112** performed by a user **106** and attributes **146** (e.g., types of hardware and software tools) recorded for these interactions **112**, UI manager **140** may be configured to determine a user profile **144** associate with the user **106**. For example, when the user **106** is determined to have used a braille keyboard and text to speech software on a computing node **104** while performing the interactions **112**, UI manager **140** determines that the user **106** is associated with a visually impaired profile.

(22) UI manager **140** may be configured to store (e.g., in memory **196**) a set of conformance rules **152** mapped to each user profile **144**. Conformance rules **152** associated with a user profile **144** define a UI design that is to be followed for UI screens **170** generated for a user **106** associated with the user profile **144**. A UI screen **170** associated with a software application **110** needs to conform to the set of conformance rules **152** defined for a particular user profile **144** associated with the user **106** using the software application **110**. Conformance rules **152** associated with a particular user profile **144** may be defined by an organization/entity that manages the software application **110** being used by the user **106**, a government institution, any other entity, or a combination thereof. For example, conformance rules **152** may include accessibility rules defined

for users **106** with disabilities. In one embodiment, conformance rules **152** may include several sets of conformance rules **152** associated with improving accessibility for disabled users, wherein each set of these conformance rules **152** is defined for a different disability associated with the respective user profile **144**. For example, a set of conformance rules **152** associated with a visually impaired user profile **144** define a UI design for UI screens **170** generated for users **106** with visual impairment. In one example, conformance rules **152** include conformance rules **152** according to Americans with Disabilities Act (ADA). In one example, a different set of conformance rules **152** is defined for each user profile **144** including, but not limited to, vision impaired profile, cognitive disability profile, seizure profile, attention deficit hyperactivity disorder (ADHD) profile, blind profile, and impaired motor skills profile.

(23) In an additional or alternative embodiment, conformance rules **152** may include data security conformance rules **152** associated with user profiles **144** indicating a degree of data security associated with a user **106**. For example, conformance rules **152** associated with a user profile **144** relating to a corporate user may define a UI design with relatively reduced security features (e.g., multi-factor authentication, security questions etc.). On the other hand, conformance rules **152** associated with a user profile **144** relating to a student user may define a UI design with relatively advanced security features.

(24) UI manager **140** may be configured to generate UI screens **170** for a user **106** in a particular UI environment **108** based on the user profile **144** determined for the user **106** as described above and the conformance rules **152** mapped to the user profile **144**. As described above, example UI environments **108** may include a web environment or a virtual environment (e.g., metaverse environment). Once a user profile **144** associated with a particular user **106** has been determined, UI manager **140** monitors for interactions **112** requested by the user **106**. For example, the user **106** may initiate a login procedure on a web server to access a social media platform in a web environment. UI manager **140** may be configured to determine the UI environment **108** associated with a request for an interaction **112**. For example, when the user **106** uses a web browser to initiate the interaction **112**, UI manager **140** determines that the UI environment **108** associated with the requested interaction **112** is a web environment. In another example, when the user **106** uses VR glasses to initiate the interaction **112** on a metaverse application, UI manager **140** determines that the UI environment **108** associated with the requested interaction **112** is a metaverse environment.

(25) In response to detecting that the user **106** has requested to perform an interaction **112** in a particular UI environment **108**, UI manager **140** obtains a UI model **148** associated with the UI environment **108**. In this context, UI manager **140** may store (e.g., in memory **196**) a UI model **148** for each UI environment **108**. For example, memory **196** may store a first UI model **148** associated with a web environment and may store a second UI model **148** associated with a virtual environment. Each UI model **148** includes a plurality of UI elements **150** that may be used to build a UI screen **170** for a particular software application **110** in the respective UI environment **108**. For example, a UI model **148** associated with a web environment may include a plurality of UI elements **150** providing accessibility features for users **106** with different types of disabilities. In this context, example UI elements **150** may include enlarged text for visually impaired users, color contrasted text for colorblind users, and closed captioning for hearing impaired users. Another UI model **148** may include example UI elements **150** such as multi-factor authentication screens, biometric verification screens and other UI screens implementing advanced data security features for high-risk users. Similarly, another UI model **148** associated with a virtual environment may include a plurality of similar UI elements **150** providing accessibility features for users **106** with different types of disabilities.

(26) To generate a UI screen **170** for the user **106**, UI manager **140** may be configured to select UI elements **150** from the obtained UI model **148** that conform to at least one conformance rule **152** defined for the user profile **144** associated with the user **106**. For example, a set of UI elements **150** included in the UI model **148** may be in conformance with a corresponding set of conformance

rules **152** defined for a particular user profile **144** associated with a particular disability. Upon determining that the user **106** who requested to perform the interaction **112** is associated with the particular user profile associated with the particular disability, UI manager **140** may select the set of UI elements **150** from the UI model **148** to generate a UI screen **170** for the user **106**. In another example, a second set of UI elements **150** from the UI model **148** may be in conformance with a corresponding set of conformance rules **152** defined for a particular user profile **144** associated with users **106** vulnerable to cyber-attacks. Upon determining that the user **106** who requested to perform the interaction **112** is associated with the particular user profile associated a higher vulnerability of cyber-attacks, UI manager **140** may select the second set of UI elements **150** from the UI model **148** to generate a UI screen **170** for the user **106**.
(27) In one or more embodiments, UI manager **140** may be configured to use a machine learning model to perform one or more operations described in this disclosure including determining a user profile **144** of a user **106** and generating a UI screen **170** based on conformance rules **152** defined for the determined user profile **144**. For example, to determine a user profile **144** of a user **106**, machine learning model **142** may be trained based on attributes **146** associated with one or more hardware devices, one or more software applications, or a combination thereof that are known to be associated with particular user profiles **144**.
(28) It may be noted that while embodiments of the present disclosure are described in the context of user profiles **144** associated with disabilities, a person having ordinary skill in the art may appreciate that the discussed techniques generally apply to determining any type of user profile **144** and generating UI screens **170** based on conformance rules **152** defined for the user profile **144**.
(29) FIG. **2** illustrates a flowchart of an example method **200** for generating a UI screen for a user **106**, in accordance an embodiment of the present disclosure. Method **200** may be performed by the UI manager **140** shown in FIG. **1**.
(30) At operation **202**, UI manager **140** monitors a plurality of interactions **112** performed by a first user **106** in a computing infrastructure **102**.
(31) As described above, UI manager **140** may be configured to determine a user profile **144** associated with a user **106** based on interactions **112** performed by the user **106** in the computing infrastructure **102**. Each user profile **144** may be associated with a particular type of users **106**. For example, a user profile **144** may be associated with users having a particular type of disability. In this example, user profiles **144** may include, but are not limited to, vision impaired profile, cognitive disability profile, seizure profile, attention deficit hyperactivity disorder (ADHD) profile, blind profile, and impaired motor skills profile. Other example user profiles **144** may include corporate user profile and individual user profile. UI manager **140** may be configured to determine a user profile **144** associated with a particular user **106** based on monitoring a plurality of interactions **112** performed by the user **106** in the computing infrastructure **102**. An interaction performed by a user **106** may include the user **106** accessing a software application **110** (e.g., a web application, metaverse application etc.) to perform a particular task such as posting on a social media platform, sending an email message, transfer data files to other users or entities, purchase event tickets and the like. For example, the user **106** may log into a social media platform running on a server as a web application or a virtual application to consume content or to submit content.
(32) At operation **204**, UI manager **140** records a plurality of attributes **146** associated with one or more hardware devices (e.g., computing nodes **104**), one or more software applications **110**, or a combination thereof used by the first user to perform the plurality of interactions **112**, wherein the plurality of attributes **146** are indicative of one or more characteristics associated with the first user **106**.
(33) At operation **206**, based on the recorded attributes **146**, UI manager **140** determines a first user profile **144** applicable to the first user **106**.
(34) As described above, monitoring interactions **112** performed by the particular user **106** may include recording a plurality of attributes **146** associated with one or more hardware devices, one or

more software applications, or a combination thereof used by the user **106** while performing the monitored interactions **112**. The attributes **146** associated with interactions **112** monitored for the user **106** may indicate one or more characteristics associated with the user **106**. In one embodiment, attributes **146** may include a type of one or more hardware devices (e.g., input devices, output devices etc.), a type of one or more software applications, or a combination thereof used by the user **106** while performing the monitored interactions **112**. In this context, the type of hardware devices and/or software applications used by a particular user **106** may indicative of a particular type of disability associated with the user **106**, and thus may indicate a particular user profile **144** associated with the particular disability. For example, a user **106** having a particular type of disability may use specific hardware devices and/or specific software tools that are designed to help the users **106** having the particular type of disability to perform interactions **112** in the computing infrastructure **102**. For example, a vision impaired user may use a braille keyboard and/or a particular software plugin that converts voice to text and vice versa to enable the user **106** to issue voice commands. Other examples of assistive hardware devices and software tools that may be used by users **106** with different types of disabilities may include, but are not limited to, screen readers, adaptive keyboards, alternative input devices, braille displays, electronic screen magnifiers, augmentative and alternative communication software, dictation software, optical character recognition (OCR) software, screen magnification software and text to speech software. Each of these assistive hardware devices and software tools help users **106** with specific types of disabilities and thus are indicative of the type of disability associated with the user **106** using such devices and software.
(35) In another example, attributes **146** may indicate whether a user **106** is employed by a large organization or entity. For example, if a user **106** uses a high-end computer and/or uses an operating system or software tools typically used by large corporations, this is indicative of the user **106** being part of a large organization or entity. On the other hand, when a user **106** uses a low-end computer and/or uses home versions or student versions of an operating system of other software tools, this is indicative of the user **106** a low-end user.
(36) In one or more embodiments, UI manager **140** may be configured to monitor one or more computing nodes **104** used by a user **106** to perform a plurality of interactions **112** and identify the particular hardware devices and software applications used on the computing nodes **104** to perform the interactions **112**. Based on monitoring a plurality of interactions **112** performed by a user **106** and attributes **146** (e.g., types of hardware and software tools) recorded for these interactions **112**, UI manager **140** may be configured to determine a user profile **144** associate with the user **106**. For example, when the user **106** is determined to have used a braille keyboard and text to speech software on a computing node **104** while performing the interactions **112**, UI manager **140** determines that the user **106** is associated with a visually impaired profile.
(37) At operation **208**, UI manager **140** monitors for a request for an interaction **112** made by the first user **106**. At operation **210**, in response to detecting that the first user **106** has requested to perform a first interaction **112** in a first UI environment **108**, method **200** proceeds to operation **212**, where UI manager **140** obtains a first UI model **148** associated with the first UI environment **108**.
(38) As described above, once a user profile **144** associated with a particular user **106** has been determined, UI manager **140** monitors for interactions **112** requested by the user **106**. For example, the user **106** may initiate a login procedure on a web server to access a social media platform in a web environment. UI manager **140** may be configured to determine the UI environment **108** associated with a request for an interaction **112**. For example, when the user **106** uses a web browser to initiate the interaction **112**, UI manager **140** determines that the UI environment **108** associated with the requested interaction **112** is a web environment. In another example, when the user **106** uses VR glasses to initiate the interaction **112** on a metaverse application, UI manager **140** determines that the UI environment **108** associated with the requested interaction **112** is a

metaverse environment.
(39) In response to detecting that the user **106** has requested to perform an interaction **112** in a particular UI environment **108**, UI manager **140** obtains a UI model **148** associated with the UI environment **108**. In this context, UI manager **140** may store (e.g., in memory **196**) a UI model **148** for each UI environment **108**. For example, memory **196** may store a first UI model **148** associated with a web environment and may store a second UI model **148** associated with a virtual environment. Each UI model **148** includes a plurality of UI elements **150** that may be used to build a UI screen **170** for a particular software application **110** in the respective UI environment **108**. For example, a UI model **148** associated with a web environment may include a plurality of UI elements **150** providing accessibility features for users **106** with different types of disabilities. In this context, example UI elements **150** may include enlarged text for visually impaired users, color contrasted text for colorblind users, and closed captioning for hearing impaired users. Another UI model **148** may include example UI elements **150** such as multi-factor authentication screens, biometric verification screens and other UI screens implementing advanced data security features for high-risk users. Similarly, another UI model **148** associated with a virtual environment may include a plurality of similar UI elements **150** providing accessibility features for users **106** with different types of disabilities.
(40) At operation **214**, UI manager **140** obtains a first set of conformance rules **152** associated with the first user profile **144**.
(41) As described above, UI manager **140** may be configured to store (e.g., in memory **196**) a set of conformance rules **152** mapped to each user profile **144**. Conformance rules **152** associated with a user profile **144** define a UI design that is to be followed for UI screens **170** generated for a user **106** associated with the user profile **144**. A UI screen **170** associated with a software application **110** needs to conform to the set of conformance rules **152** defined for a particular user profile **144** associated with the user **106** using the software application **110**. Conformance rules **152** associated with a particular user profile **144** may be defined by an organization/entity that manages the software application **110** being used by the user **106**, a government institution, any other entity, or a combination thereof. For example, conformance rules **152** may include accessibility rules defined for users **106** with disabilities. In one embodiment, conformance rules **152** may include several sets of conformance rules **152** associated with improving accessibility for disabled users, wherein each set of these conformance rules **152** is defined for a different disability associated with the respective user profile **144**. For example, a set of conformance rules **152** associated with a visually impaired user profile **144** define a UI design for UI screens **170** generated for users **106** with visual impairment. In one example, conformance rules **152** include conformance rules **152** according to Americans with Disabilities Act (ADA). In one example, a different set of conformance rules **152** is defined for each user profile **144** including, but not limited to, vision impaired profile, cognitive disability profile, seizure profile, attention deficit hyperactivity disorder (ADHD) profile, blind profile, and impaired motor skills profile.
(42) In an additional or alternative embodiment, conformance rules **152** may include data security conformance rules **152** associated with user profiles **144** indicating a degree of data security associated with a user **106**. For example, conformance rules **152** associated with a user profile **144** relating to a corporate user may define a UI design with relatively reduced security features (e.g., multi-factor authentication, security questions etc.). On the other hand, conformance rules **152** associated with a user profile **144** relating to a student user may define a UI design with relatively advanced security features.
(43) At operation **216**, UI manager **140** generates a first user interface screen **170** for the first UI environment **108** based on the first UI model **148** and the first set of conformance rules **152**, wherein generating the first UI screen **170** comprises selecting one or more UI elements **150** of the UI model **148** based on the first set of conformance rules **152**, wherein each UI element **150** conforms to one or more conformance rules **152**.

(44) As described above, to generate a UI screen **170** for the user **106**, UI manager **140** may be configured to select UI elements **150** from the obtained UI model **148** that conform to at least one conformance rule **152** defined for the user profile **144** associated with the user **106**. For example, a set of UI elements **150** included in the UI model **148** may be in conformance with a corresponding set of conformance rules **152** defined for a particular user profile **144** associated with a particular disability. Upon determining that the user **106** who requested to perform the interaction **112** is associated with the particular user profile associated with the particular disability, UI manager **140** may select the set of UI elements **150** from the UI model **148** to generate a UI screen **170** for the user **106**. In another example, a second set of UI elements **150** from the UI model **148** may be in conformance with a corresponding set of conformance rules **152** defined for a particular user profile **144** associated with users **106** vulnerable to cyber-attacks. Upon determining that the user **106** who requested to perform the interaction **112** is associated with the particular user profile associated a higher vulnerability of cyber-attacks, UI manager **140** may select the second set of UI elements **150** from the UI model **148** to generate a UI screen **170** for the user **106**.
(45) At operation **218**, UI manager **140** renders the first UI screen **170** in the first UI environment **108** on a first user device (e.g., computing node **104**) of the first user **106**.
(46) While several embodiments have been provided in the present disclosure, it should be understood that the disclosed systems and methods might be embodied in many other specific forms without departing from the spirit or scope of the present disclosure. The present examples are to be considered as illustrative and not restrictive, and the intention is not to be limited to the details given herein. For example, the various elements or components may be combined or integrated in another system or certain features may be omitted, or not implemented.
(47) In addition, techniques, systems, subsystems, and methods described and illustrated in the various embodiments as discrete or separate may be combined or integrated with other systems, modules, techniques, or methods without departing from the scope of the present disclosure. Other items shown or discussed as coupled or directly coupled or communicating with each other may be indirectly coupled or communicating through some interface, device, or intermediate component whether electrically, mechanically, or otherwise. Other examples of changes, substitutions, and alterations are ascertainable by one skilled in the art and could be made without departing from the spirit and scope disclosed herein.
(48) To aid the Patent Office, and any readers of any patent issued on this application in interpreting the claims appended hereto, applicants note that they do not intend any of the appended claims to invoke 35 U.S.C. § 112(f) as it exists on the date of filing hereof unless the words “means for” or “step for” are explicitly used in the particular claim.

Claims

1. A system comprising: a memory storing a plurality of user profiles and a set of conformance rules associated with each user profile; and a processor communicatively coupled to the memory and configured to: monitor a plurality of interactions performed by a first user in a computing infrastructure; record, based on the monitoring, a plurality of attributes associated with one or more hardware devices, one or more software applications, or a combination thereof used by the first user to perform the plurality of interactions, wherein the plurality of attributes are indicative of one or more characteristics associated with the first user; determine, based on the recorded plurality of attributes, a first user profile of the plurality of user profiles applicable to the first user; detect that the first user has requested to perform a first interaction in a first user interface environment; obtain a first user interface model associated with the first user interface environment; obtain a first set of conformance rules associated with the first user profile; generate a first user interface screen for the first user interface environment based on the first user interface model and the first set of conformance rules, wherein generating the first user interface screen comprises selecting one or

more user interface elements of the first user interface model based on the first set of conformance rules, wherein each user interface element conforms to one or more conformance rules; and render the first user interface screen in the first user interface environment on a first user device of the first user; wherein the plurality of user profiles comprise a set of user profiles associated with user disabilities, wherein each user profile from the set is associated with a different type of user disability; wherein the processor is further configured to: determine the plurality of attributes by determining that the first user used a hardware input device, a hardware output device, a software application, or a combination thereof designed for use by persons of a particular type of disability; and select the first user profile associated with the particular type of disability from the plurality of user profiles; wherein the first set of conformance rules comprises user interface standards defined for persons associated with the particular type of disability; and wherein the processor generates the first user interface screen by selecting from the first user interface model one or more user interface elements that conform to the user interface standards defined for persons associated with the particular type of disability.

2. The system of claim 1, wherein the plurality of attributes comprises a type of one or more hardware input devices, a type of one or more hardware output devices, a type of one or more software applications, or a combination thereof.

3. The system of claim 1, wherein the first user interface environment comprises a web environment or a virtual environment.

4. A method for generating UI screens, comprising: monitoring a plurality of interactions performed by a first user in a computing infrastructure; recording, based on the monitoring, a plurality of attributes associated with one or more hardware devices, one or more software applications, or a combination thereof used by the first user to perform the plurality of interactions, wherein the plurality of attributes are indicative of one or more characteristics associated with the first user; determining, based on the recorded plurality of attributes, a first user profile of a plurality of user profiles applicable to the first user; detecting that the first user has requested to perform a first interaction in a first user interface environment; obtaining a first user interface model associated with the first user interface environment; obtaining a first set of conformance rules associated with the first user profile, wherein a set of conformance rules is associated with each user profile; generating a first user interface screen for the first user interface environment based on the first user interface model and the first set of conformance rules, wherein generating the first user interface screen comprises selecting one or more user interface elements of the first user interface model based on the first set of conformance rules, wherein each user interface element conforms to one or more conformance rules; and rendering the first user interface screen in the first user interface environment on a first user device of the first user; wherein the plurality of user profiles comprise a set of user profiles associated with user disabilities, wherein each user profile from the set is associated with a different type of user disability; the method further comprising: determining the plurality of attributes by determining that the first user used a hardware input device, a hardware output device, a software application, or a combination thereof designed for use by persons of a particular type of disability; and selecting the first user profile associated with the particular type of disability from the plurality of user profiles; wherein the first set of conformance rules comprises user interface standards defined for persons associated with the particular type of disability; and wherein generating the first user interface screen comprises selecting from the first user interface model one or more user interface elements that conform to the user interface standards defined for persons associated with the particular type of disability.

5. The method of claim 4, wherein the plurality of attributes comprises a type of one or more hardware input devices, a type of one or more hardware output devices, a type of one or more software applications, or a combination thereof.

6. The method of claim 4, wherein the first user interface environment comprises a web environment or a virtual environment.

7. A non-transitory computer-readable medium storing instructions that when executed by a processor cause the processor to: monitor a plurality of interactions performed by a first user in a computing infrastructure; record, based on the monitoring, a plurality of attributes associated with one or more hardware devices, one or more software applications, or a combination thereof used by the first user to perform the plurality of interactions, wherein the plurality of attributes are indicative of one or more characteristics associated with the first user; determine, based on the recorded plurality of attributes, a first user profile of a plurality of user profiles applicable to the first user; detect that the first user has requested to perform a first interaction in a first user interface environment; obtain a first user interface model associated with the first user interface environment; obtain a first set of conformance rules associated with the first user profile, wherein a set of conformance rules is associated with each user profile; generate a first user interface screen for the first user interface environment based on the first user interface model and the first set of conformance rules, wherein generating the first user interface screen comprises selecting one or more user interface elements of the first user interface model based on the first set of conformance rules, wherein each user interface element conforms to one or more conformance rules; and render the first user interface screen in the first user interface environment on a first user device of the first user; wherein the plurality of user profiles comprise a set of user profiles associated with user disabilities, wherein each user profile from the set is associated with a different type of user disability; wherein the instructions further cause the processor to: determine the plurality of attributes by determining that the first user used a hardware input device, a hardware output device, a software application, or a combination thereof designed for use by persons of a particular type of disability; and select the first user profile associated with the particular type of disability from the plurality of user profiles; wherein the first set of conformance rules comprises user interface standards defined for persons associated with the particular type of disability; wherein generating the first user interface screen comprises selecting from the first user interface model one or more user interface elements that conform to the user interface standards defined for persons associated with the particular type of disability.
8. The non-transitory computer-readable medium of claim 7, wherein the plurality of attributes comprises a type of one or more hardware input devices, a type of one or more hardware output devices, a type of one or more software applications, or a combination thereof.